

LAE Future Frontiers & Final Review

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Low Altitude Economics

Week 12

Learning Objectives

By the end of this class, you should be able to:

- **Explain** how adding nodes increases possible connections without guaranteeing demand or value.
- **Evaluate** digital coordination using total system cost, reliability, oversight, and governance.
- **Assess** possible Hong Kong roles in financing, leasing, insurance, professional services, and data-enabled assurance.
- **Apply** the course viability test to integrated quantitative scenarios.
- **Develop** recommendations that remain consistent with operational, regulatory, financial, and social constraints.

The Next Horizon

The final week asks how an LAE project moves from a controlled trial to a repeatable operating system.

- A larger network may create more useful connections, but also more coordination, repositioning, infrastructure, and regulatory requirements.
- Growth should therefore follow evidence on demand, reliable capacity, unit economics, safety, and stakeholder impact.

Quick poll (2 min): Which part of the value chain is most likely to create durable value: aircraft, operations, infrastructure, software, finance, or professional services?

Concept Primer: From Corridors to Networks

A network links several origins, destinations, operators, and supporting facilities.

- More nodes create more possible connections, but only some will have sufficient demand and feasible operations.
- Network value depends on frequency, reliability, access, transfer quality, compatible systems, and customer needs.
- Cross-boundary services also require coordination concerning permissions, customs, immigration, quarantine, data, communications, liability, and emergency response.
- Empty repositioning, unbalanced demand, congestion, and duplicated infrastructure can offset network benefits.

Managerial lesson: Count possible links first, then test which links are useful, serviceable, and commercially viable.

Applied Tool: Counting Possible Connections

Undirected Connections

$$L = \frac{N(N-1)}{2}$$

- N is the number of active nodes and L is the number of possible pairs.
- If direction matters, the number of possible origin-destination links is $N(N-1)$.
- The formula counts possibilities, not scheduled routes, flights, demand, profit, or network value.
- The number of links grows quadratically with N , not exponentially.
- Each proposed link still requires demand, capacity, infrastructure, operating permission, and acceptable cost.

Back-of-the-Envelope: Possible and Viable Links

Illustrative network expansion:

Five Nodes

- Possible node pairs: $5(4)/2 = 10$
- Suppose six pairs pass demand and feasibility screening
- **Candidate links: 6**

Twenty Nodes

- Possible node pairs: $20(19)/2 = 190$
- Suppose 15% pass initial screening
- **Candidate links: about 29**

Takeaway: More nodes increase the option set, but viable service grows only where demand, capacity, regulation, and operations align.

Illustrative screening rate; not a forecast for the GBA.

Concept Primer: Digital Coordination and Governance

Digital systems can support planning, authorization, identification, traffic information, monitoring, and conflict management.

- Automation may reduce repetitive manual work and improve consistency at higher volumes.
- Operators and service providers still require defined responsibilities, qualified personnel, cybersecurity, assurance, incident response, and regulatory oversight.
- Human roles may change rather than disappear, particularly for exceptions, emergencies, system degradation, and accountability.
- Algorithmic decisions should be traceable, testable, secure, and subject to appropriate review.

Managerial lesson: Evaluate the complete socio-technical system, not simply the cost of an algorithm.

Applied Tool: Digital-Coordination Business Case

Annual Net Saving

$$\begin{aligned} \text{Net Saving} = & \text{Cost of Current Process} \\ & - \text{Cost of Digital Process} - \text{Transition Cost} \end{aligned}$$

- Include software, communications, cybersecurity, certification, human oversight, maintenance, service-provider fees, and exception handling.
- Savings depend on annual activity, reliability, integration complexity, and the amount of manual work actually displaced.
- Marginal digital cost may be low, but it is not generally zero.
- The preferred design must also meet safety, resilience, privacy, and accountability requirements.

Back-of-the-Envelope: Value of Better Risk Evidence

Illustrative assumptions: Annual premium without project-specific evidence is HK\$12.5M. Evidence collection and risk controls cost HK\$3M.

No Demonstrated Improvement

- Annual premium: HK\$12.5M
- Evidence and control cost: HK\$0
- **Total: HK\$12.5M**

Improved Risk Case

- Illustrative premium: HK\$9M
- Evidence and control cost: HK\$3M
- **Total: HK\$12M**
- Illustrative first-year saving: HK\$0.5M

Takeaway: Better evidence may support underwriting and risk management, but premium reductions are not automatic and must be evaluated against control costs.

Illustrative assumptions; not an insurance quotation or a measured Sandbox X effect.

Session 2

HK's Services Edge & Global Standard Setting

Concept Primer: Possible Hong Kong Service Roles

A 2025 Our Hong Kong Foundation projection assigns a large share of potential 2050 LAE value added to supporting services.

- Possible roles include finance, leasing, insurance, legal and professional services, certification support, data assurance, cybersecurity, research, and training.
- Hong Kong's actual advantage must be tested through capability, cost, regulation, customer demand, access to regional markets, and competition.
- Manufacturing, operations, infrastructure, and services are complementary rather than mutually exclusive layers.
- High value added does not imply high margin, low risk, or automatic market leadership.

Managerial lesson: Identify a specific service, customer problem, capability, revenue model, and defensible source of advantage.

Applied Tool: A Simple Leasing Appraisal

Annual Lessor Cash Flow

$CF_L = \text{Lease Receipts} - \text{Financing Cost} - \text{Maintenance} - \text{Insurance} - \text{Expected C}$

Investment Value

$$NPV = -\text{Purchase Price} + \sum_{t=1}^T \frac{CF_{L,t}}{(1+r)^t} + \frac{\text{Residual Value}_T}{(1+r)^T}$$

- Depreciation is an accounting allocation; residual value and cash expenditures belong in the cash-flow appraisal.
- The lessor retains asset-value, financing, technology, maintenance, repossession, legal, and lessee-credit risks.
- Lease receipts are contractual, not risk-free or guaranteed in economic terms.

Back-of-the-Envelope: Risk Does Not Disappear

Illustrative monthly scenario: Lease payment HK\$300k.

Operator

- Revenue after disruption: HK\$250k
- Lease payment: HK\$300k
- Other operating costs omitted
- **Shortfall before other costs: HK\$50k**
- Bears demand and operating risk

Lessor

- Contractual lease receipt: HK\$300k
- Financing and asset-management cost: HK\$150k
- Cash flow before credit loss: HK\$150k
- **If the lessee defaults, receipt and asset value are at risk**

Takeaway: Leasing reallocates risks between parties; it does not create a risk-free margin.

Illustrative assumptions; excludes maintenance, insurance, tax, residual value, and recovery cost.

From Local Evidence to Transferable Capability

Controlled trials can generate evidence on aircraft performance, communications, navigation, weather, emergency response, infrastructure, simultaneous operations, and user feedback.

- Evidence may support local policy development, operating procedures, assurance services, and commercial refinement.
- Transfer to another jurisdiction is not automatic because rules, environments, aircraft, routes, and acceptance conditions differ.
- A credible exportable capability may include testing methods, data governance, assurance, training, technical integration, or professional advice.
- Claims of a global “gold standard” require external validation, reproducibility, interoperability, and recognition by relevant authorities.

Managerial lesson: Export validated methods and capabilities, not an assumption that one local rulebook will be universally adopted.

Session 3

Quantitative Applications & Final Exam Prep

The Final Exam: A Five-Step Method

For an integrated LAE scenario:

1. **Define the decision:** What is being chosen, for whom, over what period, and compared with what alternative?
2. **Find served activity:** Compare customer demand with reliable capacity.
3. **Calculate viability:** Use price, variable cost, served activity, fixed cost, and any specified charges or payments.
4. **Map each constraint once:** Show whether it changes demand, capacity, price, cost, timing, or risk.
5. **Recommend conditionally:** Address safety, regulation, sustainability, distribution, uncertainty, and the evidence needed to proceed.

The aim is not to force every course tool into every answer. Use only the tools that clarify the decision.

Applied Tool: The Viability Test

Step 1: How Much Can We Actually Serve?

$$\text{Served Activity} = \min\{\text{Customer Demand}, \text{Reliable Capacity}\}$$

Step 2: Does the Service Cover Its Costs?

$$\text{Operating Profit} = (\text{Price} - \text{Variable Cost}) \times \text{Served Activity} - \text{Fixed Cost}$$

- Add any specified tax, fee, subsidy, or committed payment in the correct place.
- Recalculate served activity when a shock changes demand or capacity.
- Then discuss regulation, safety, sustainability, and distributional effects.

Quantitative Toolkit Recap

Choose the tool that matches the question:

- **Demand:** generalized cost, choice share, elasticity, and market funnel.
- **Capacity:** reliable flights, seats or payload, operating days, and bottlenecks.
- **Economics:** contribution, breakeven activity, unit cost, cash flow, and NPV.
- **Uncertainty:** scenarios, sensitivity analysis, expected value, probability of loss, and liquidity need.
- **Allocation:** queueing, reservations, priority rules, congestion charges, and auctions.
- **Impact:** life-cycle emissions, noise exposure, privacy controls, stakeholder distribution, and KPIs.

Start with the decision and units. Do not begin with an all-purpose “super-formula.”

Back-of-the-Envelope: A Capacity and Cost Shock

Base data: Demand = 200 flights/day; reliable capacity after disruption = 160; price = HK\$500; variable cost = HK\$100/flight; fixed cost = HK\$40,000/day; specified noise charge = HK\$50/flight.

Step 1: Served Activity

$$\min\{200, 160\} = 160$$

Step 2: Contribution

$$(500 - 100 - 50) \times 160 = \text{HK\$56,000}$$

Step 3: Operating Profit

$$56,000 - 40,000 = \text{HK\$16,000/day}$$

- The service remains profitable under the stated assumptions.
- Compare mitigation cost with the expected reduction in the charge and other benefits.

Do not recommend visual masking as a noise remedy. Privacy and noise require different controls.

Illustrative exam assumptions; not a current Hong Kong noise charge or capacity rule.

The Ultimate Goal: Consistent Judgment

A strong answer combines correct calculation with a recommendation that respects the operating context.

- Keep units consistent and distinguish demand, capacity, served activity, flights, seats, and passenger-trips.
- Do not apply the same constraint twice through separate “haircuts.”
- Do not assume that Sandbox X bypasses legal or safety requirements; it supports controlled testing and evidence collection.
- Do not assume a market mechanism is appropriate where safety or predefined public-service priority governs access.
- State which assumptions are provided, which are sourced, and which are analytical scenarios.
- Conclude with the condition under which the project should proceed, pause, redesign, or stop.

Conclusion: From Analysis to Decision

The Low Altitude Economy remains an emerging field in which technology, regulation, business models, and public expectations continue to develop.

- Good analysis separates evidence from assumptions.
- Good modelling connects demand, reliable capacity, cost, finance, risk, and impact without double counting.
- Good strategy uses staged commitments and evidence-based decision gates.
- Good policy considers safety, efficiency, innovation, competition, accessibility, privacy, environmental impact, and distribution.

The objective is not to predict the future with certainty, but to make better decisions as evidence develops.

End of Course

Final Reminders:

- *Review all past reading bundles and calculation frameworks.*
- *Final Exam logistics and timing.*

Questions for the Final Exam?

(Final 15 minutes reserved for extended Q&A)